Comparing Satisfaction Between International and Irish Masters Taught Students in Irish Higher Education

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Abstract

The aim of this study was to investigate that if there are a different levels of overall satisfaction for the International and the Irish Masters Taught students in the Higher Education sector in the Ireland and also to explain the reasons for any differences. The analysis was undertaken using the StudentSurvey.ie survey data as the primary data set in the analysis,it comprising 7,188 Masters Taught students in the year 2023, and also 5,745 students in the year 2019 to look at the changes over the time. The primary analytical method that was used was binary logistic regression, complemented by the ordinal regression models and additional diagnostics.The overall student satisfaction was not related to whether a student was the Irish or international. In all the models, there was no such evidence that being an Irish student or an international student, in isolation did predict if a student reported a positive satisfaction. However when the aspects of the study mode and the field of study were taken in together, the relationship did changed. The analysis showed that any difference related to the domicile was dependent on the academic context in which students were studying, the direction and magnitude of the relationship differed according to the different combinations of the study mode and the field of study.when Compared to the year 2019, it was found that the satisfaction gap between the Irish and the international students had closed. Further in the modelling indicated it was more than a descriptive pattern. The overall pattern of the satisfaction among the Irish students was generally stable over the two years of the surveys, while for the international students there was a statistically significant increase, after controlling for the study mode and the field of study. The Masters Taught population also saw the significant change in its makeup over the same timeframe, with the number of the international students rising from just over one-third of a Masters Taught cohort in 2019, to just over one-half in 2023. This trend helps to illustrate the evolving nature of the postgraduate education in Ireland.

Domicile alone could not account for the differences in the overall student satisfaction. Rather, they arise through the interaction between the domicile and the students academic circumstances (study mode and field of study). These patterns have been identified as an existing and the satisfaction gap closing from the year 2019 to 2023, but in further data on the survey is not available to determine the reasons for these changes.

1 Introduction

Between the year 2019 and year 2023, the number of students in Masters Taught programme changed a lot .There were approximately 5,745 students in the year 2019. The Irish-domiciled(students whose permanent address before starting the programme was in Ireland, including Northern Ireland) would cover around 64.3% of the total(3,695 students)and remaining 35.7% (2,050 students) were the international students(students whose permanent address before starting the programme was outside Ireland). Around 7,188 students were there in the year 2023. Also, the number of students in each of the Domicile was also changed. The international students were increased upto 3,740 covering 52% of total Masters Taught program, meanwhile the Irish-domiciled students were 3,448 and about 48% of the total. During the 4 years period,the number of International students have grown from one third to majority in Master Taught program. This sudden change was not due to the overall growth in enrolments. The overall student population grew by around 25%, with a growth of about 82% in the numbers of international students and a decrease of about 6.7% in the numbers of students who are Irish-domiciled. Together, these numbers indicate an overall trend toward a change in the makeup of the Master Taught program.During the years, however the international students became the largest group of students.

It is a system in the Ireland, where the quality assurance will be kept in a check, and the Ireland higher education institution will at the same time be competing with the other countries to attract and also retain the students. This has implications for the satisfaction levels between the Irish and the international students, which is a factor to be taken into consideration. This is not just only about one institute at all . This is also relevant for the whole countrys rules on the Higher education. The same comparison between the two groups of the people over the single factor of the domicile would be too simple. The field of study or course that students have taken and the type of enrolment (full time or part time) also varies. These factors may affect the feel and also university life in the Ireland. This is why the domicile is considered jointly with the study mode and the field of study, rather than as a standalone factor. The period of the year between 2019-2023 is also an important time, as it was during the COVID-19 pandemic and in the midst of a shift in the economic conditions including rising cost of living and a dramatic shift in the graduate opportunities. It is my guess that both of these could have an impact on the students overall satisfaction with their educational experience in this country.

The analysis was done based on the dataset of the year 2019 and year 2023 from StudentSurvey.ie national survey. A total of 39,403 first year undergraduates, final year undergraduates and the taught postgraduates responded to this national survey in 2023. The data used in this study has been limited to only Mater Taught students. So the finally it consists of 5,745 students from the 2019 dataset and 7,188 students from the 2023 dataset. In the survey, the overall satisfaction questions was used to measure the students satisfaction and the students satisfaction had four response categories namely, Poor, Fair, Good and Excellent. All these responses were in further regrouped into two groups to make them “analysis appropriate”. It was determined that by this classification that the Good and Excellent were classified as the positive satisfaction and Poor and Fair were classified as the negative satisfaction and this classification made it a possible to use the binary logistic regression which was used to exam-

ine how the effect of students domicile on the likelihood of reporting the positive satisfaction changed as a function of the combination of the study mode and the field of study.

This study is focused on the research question that is:"Do international Masters Taught students in Irish higher education report different levels of overall satisfaction compared to their domestic counterparts, and what factors explain any differences that exist?" This study is particularly focused on the Master Taught students in Ireland. The research is targeted at the Masters Taught group of students, so as to keep the research clear and consistent in the analysis.

The first thing that to remember when reading the results of these is that they are being considered alongside it, rather than as a replacement for, the findings from the primary 2023 model.This should not be interpreted as being contradictory as the 2 models are based upon answering the different questions. In the primary model, the overall satisfaction was not statistically significantly associated with the domicile ($p = 0.945$),a result discussed in detail in a Section 5.2 alongside the Table 2. But this does not mean that the domicile did not have an impact on it.Instead that the model indicated that the effect of domicile was dependent upon the study mode and the field of study, and was not necessarily the same for all the students. The model specification used here is different and addresses a different question than is examined in the analysis in this paper. This has the effect of complementing rather than replacing the main model in the findings. The difference in the overall satisfaction between the Irish and the international students had already been identified in the detailed results and it was noted that this gap became less over the time. In the year 2019,the students satisfaction overall was 83.0% for the Irish students and 77.9% for the international students, a gap of 5.1 percentage points between the two groups. In the year 2023 both groups reported the same level of the positive satisfaction (81.5%), indicating that the difference had been eliminated. The interaction model reveals that this change was not just a random occurrence in the data but it is statistically significant.

The dissertation is being divided into six chapters, with each chapter following the other. In chapter 2, the background to the study is given in the introduction. It presents the StudentSurvey.ie data set, describing the way in which it is being implemented in the study, who will be running it and how students are being asked to take part in the survey. It also detailed the data that would be being collected for this research from the Irish Social Science Data Archive (ISSDA). The methodological approach used for the study is described in the Chapter 3, where it is explained how the data from the original survey were prepared for the analysis. It provides an explanation of the variables selected for the analysis, and why the focus is on the Masters Taught students only, which is where the meaningful comparisons between the Irish and the international students can be made, and what approaches is being taken to handling any missing data in the study. It also looks at the extent to which those who did not respond to the satisfaction question still finished the rest of the survey. The exploratory data analysis, which is presented in Chapter 4, is used to examine the relationships between the variables (domicile, study mode, field of study and overall satisfaction) using detailed statistics, figures and cross-tabulations before any statistical modelling is being undertaken. The modelling process is then described in chapter 5, starting with the binary logistic regression, to be followed by the main three-way interaction model, to be checked for adequacy with diagnostic checks, and to be described in additional models for comparison.

In the last chapter, the key results of the study are being summarized, the limitations of the study are being discussed and the potential for future research is explored

2 Background

The data presented in this study has been sourced from the Irish Survey of Student Engagement (ISSE), also known as the StudentSurvey.ie, it is an annual national survey carried out in all the higher education institutions in the State of Ireland. It is a survey to collect the students opinions on their experiences within higher education and is open to all the first year undergraduate students, final year undergraduate students and the taught postgraduate students. All eligible students at a participating school are being invited to participate on a volunteer basis rather than a few sample of the students. The survey is organised in collaboration with the Higher Education Authority (HEA), the Irish Universities Association (IUA), the Technological Higher Education Association (THEA) and the Union of Students in Ireland (USI). This contains a wide variety of topics related to student engagement which will including teaching and learning, academic support, wider student experience, satisfaction with a variety of aspects of the university life and students overall assessment of the course(Erskine, Harmon and O'Regan, 2023). The features are leading to the nationally comparable information on the experiences of students in Irish HEIs (StudentSurvey.ie, n.d.).

The year 2023 StudentSurvey.ie dataset which is used in this study was obtained under the licence from the Irish Social Science Data Archive (ISSDA), which will provides an access to the research datasets for the academic purposes. The full data set includes the answers from 39,403 students from 21 Higher Education Institutions (HEIs) across Ireland and it contains 153 variables that is capturing various aspects of the student journey from the HEI. But only a few of these variables were applicable to the present study, which includes students domicile, Field of study ,study mode , the overall satisfaction and some selected demographic parameters. The initial Data collection took place during the month of February and March, with the overall response rate of 25.7% (StudentSurvey.ie, 2023) of the number of eligible students. As the participation in the survey was ment to be done voluntarly, this relatively modest response rate is being acknowledged as a limitation as well. It is very likely that there may have been some differences between the students who did and did not respond to the survey, as only about one-quarters of the eligible students responded. The students who were either particularly pleased or displeased with their experience may be more likely to complete this survey; this may have affected the findings and their representativeness of the much broader student population.

This years dataset was being supplemented by the year 2019 StudentSurvey.ie dataset to give a pre-pandemic perspective or the comparison. The comparison of these two years can consider if the reported satisfaction of the Masters Taught students was different in a pre-pandemic period compared to a post-pandemic period, but can not isolate the impact the pandemic had from other changes in these two years which is being conisdered, including changes in the overall economic landscape, or changes in student body. Both the datasets are derived from the same national survey, but there are some differences in the naming of the variables, and some differences in the labelling of the response items.These differences

are been handled during the data preparation process and are discussed in more detail in the following methodology chapter. After the data had been prepared, the analysis is mainly focused only on the students who enrolled in the Masters Taught programmes. This led to a sample size of 5,745 students in the year 2019 and 7,188 students from the year 2023 survey. Within these groups, 2050 students, in the year 2019, and 3740 students in the year 2023, were identified as the international students (see Figure 1).

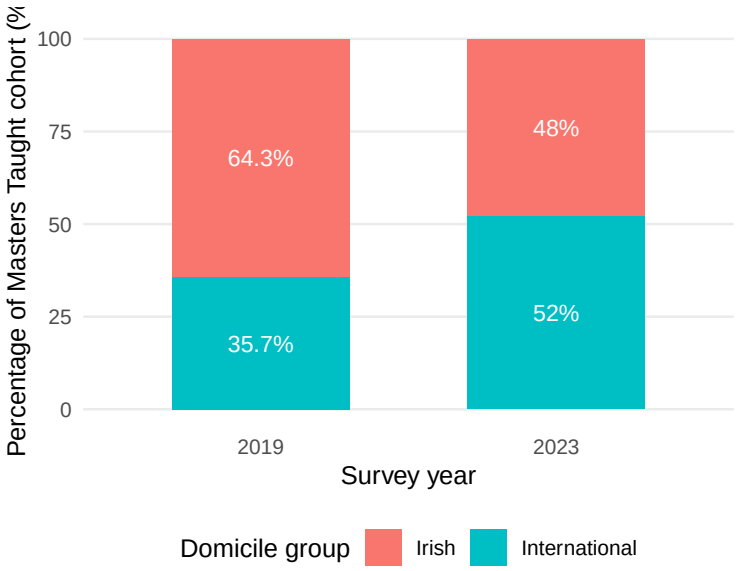


Figure 1: Composition of the Masters Taught student cohort by domicile group, 2019 and 2023.

The higher percentage of the international students in the latter survey is a reflection of the ongoing increase in the international student numbers in the Irish higher education sector, during this time.

The StudentSurvey.ie should also be acknowledged as a repeated cross sectional survey, rather than a longitudinal study. The students who took part in the year 2019 survey are not the same as those who completed the year 2023 survey, as the surveys are conducted annually on a different group of the eligible students. Hence any comparisons between the two data sets reflect the differences between the two groups, rather than changes over the time for the same students. This differences is significant in the interpretation of the findings because it would be possible to detect the patterns and the differences between the two survey years that are been taken , but it would not be possible to say that whether the individual students satisfaction level went up or down over the course of the study.

A small number of the students in the Masters Taught programmes may have been participated in both the year 2019 and the year 2023 surveys if their studies spanned an extended period covering both the survey years. However due to the lack of a unique student identifier in the dataset, it cannot be verified and quantified that whether there is any such overlap. For this analysis the two sample data sets were thus considered independent.

This study used the overall rating of the students educational experiences as main the out-

come variable. Students were asked to rate their experience on a 4 point ordered response scale: Poor, Fair, Good and Excellent. The responses are each individual students own view of their overall experience and not of the institution or the specific programme in the institute. Given this, the unit of analysis for this study is the individual student respondent and each record in the data set represents one students response to the survey. The data presented in this dissertation came from the StudentSurvey.ie data set, which comprised students from various levels of the study, but the analysis was limited to the Masters Taught students only. The reason for the choice of this group and the process used for creating the analytical dataset is being explained in the following section in the Methodology chapter.

3 Methodology

3.1 Data Access and Loading

The analysis was done using R version 4.5.1 and the RStudio. ISSDA provided the year 2023 StudentSurvey.ie data as a CSV file that was imported with the base R function read.csv(). The dataset from the year 2019 was provided as an SPSS file (.sav) and loaded using the function read_sav() from the haven package (this keeps the variable and value labels that were in the original file). The choice of import function in each case simply reflected the file format in which ISSDA supplied the data, and did not signify any difference in the analytical approach.

The raw data used in the study were not modified at any stage of the study. Data cleaning, recoding, variable selection and subsetting were all being done after the data was imported into the R, which meant that it was not changing the files and the original file could always be referred to if necessary.

3.2 Subsetting to Masters Taught Students

In the present study, the analysis was limited to the students enrolled in the Masters Taught type of programme so that it provided an equal opportunity of comparing the domestic and the international students in the same programme type. For the international taught postgraduate students, 94.5% were being enrolled on the Masters Taught programmes while 69.5% of the Irish students were being enrolled on the Masters Taught programmes. The remaining Irish taught postgraduate students were distributed all across programmes like the Higher Diploma, Postgraduate Certificate, and Postgraduate Diploma, while only a small proportion of the international students were enrolled in these programme types. However this analysis excludes the programme type as a source of the variation between the two domicile groups as only the Masters Taught students are being included. A single analysis of all the taught postgraduate programmes would have allowed any difference in the overall satisfaction for their programme to be due to the programme type but not due to differences arising from the domicile. Focusing on one programme type will allow the direct comparison

of the domestic and the international students, and minimize the risk of the results being affected by the different distribution of students in the different postgraduate programmes.

3.3 Variable Selection

The focus of this study is the comparison between the Irish and the international students, hence the primary explanatory variable the domicile, has been used in all of the models selected.Other factors considered as demographic control variables to account for the differences in the make-up of students based on variables such as the Age and the gender were also retained in the analysis.The variable Residence was included in the original model but was then dropped from the final model. The likelihood ratio test was used to check if the addition of the variable residence increased the overall model fit when compared to a model without the residence. The final test results a likelihood ratio statistic with 5 degrees of freedom and a p-value of 0.842, it is indicating that the residence did not offer evidence of improvement in the model, once the other variables had been accounted for. Therefore, residence was not included in the final model because there was no benefit in including it in the analysis as it would only add the complexity to the model.The Study mode and also the field of study were included because in the previous exploratory analyses, these both variables were found to be related to the overall satisfaction and both variables are a part of the interaction terms which are examined in with the first model.

3.4 ISCED Grouping

An additional preparation was necessary before the two survey years were merged into one analytical dataset.One of the variables in the field of study had to be prepared before the two survey years could be merged into one analytical data set. The subject classifications were similar for both the datasets, but there were some significant differences in the labels of the subject categories.For one, the year 2023 dataset has an 'and' separated by a comma that is Business, administration, and law; but the year 2019 dataset has an 'and' separated by none that is Business, administration and law. The differences were being standardised during the data cleaning process, so that the equivalent categories were matched properly between the two datasets. There were eleven categories in the original field of study variable, but for analysis it was recoded into four major categories. There were two principal reasons for making this decision. First reason was due to the some of the categories contained very few observations, particularly Generic Programmes (n = 3) and Agriculture (n = 15) in the year 2019 dataset. If these programmes are kept as distinct categories, then, due to the small number of observations in the dataset, the estimates from the logistic regression may be unstable. The second reason was with the combination of the categories resulting in a more manageable and interpretable interaction structure in the final model that is especially when the field of study was analysed in addition with domicile and study mode. The four groups that have been analyzed are STEM, Business, Health and Rest. The STEM category was made up of courses like Engineering, Manufacturing and Construction; Information and Communication Technologies; and Natural Sciences, Mathematics and Statistics. Business is

combination of Business, administration and law, while Health included Health and welfare. All other subjects not classified as either of the above, such as Education, Arts and Humanities, Social Sciences, Services, Generic Programmes and Agriculture, were grouped together in the Rest category.

3.5 Missing Data

In Prior to conducting the main analysis, the pattern of missing data in the outcome variable was to be explored to determine if the missing responses could be considered Missing Completely At Random (MCAR). In the MCAR case, the probability that a students overall satisfaction response is missing is not related to the observed characteristics of the student, or the missing satisfaction value itself. If this assumption is true, then only the complete cases should be analysed, because the results obtained in this way are unbiased, and no need to impute the missing values at all.

Several of the student characteristics were compared with the percentage of the students who responded to the item. Figure 2 shows that, for the year 2023, the percentage of missing data was 25.8% for the international students and 27.6% for the Irish students. The same trends were seen throughout the study mode, with the missing percentages of 26.5% for the full-time students and 27.3% for the part-time students. The rates of missingness were virtually the same for the age groups, with 27.1% of students aged 23 years and under missing and 26.6% of those who aged 24 years and over missing. Data on gender also did not vary significantly, with 26.5% of the females missing data and 26.9% of the males missing data, and 25.6% for those labeled undeclared students. The year 2019 data showed a similar trend that the missing rates for the Irish students was 27.0%, and for the international students was 27.5%.

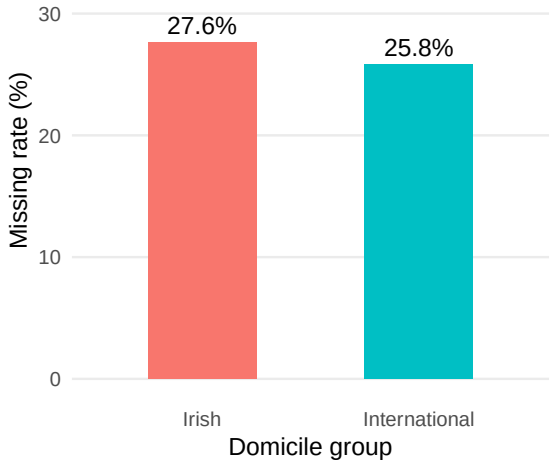


Figure 2: Missing rates for the overall evaluation outcome by domicile group, 2023.

The regularity of the missing data rates for the variables observed indicates that the non-response was not systematically related to the variables of the domicile, study mode, age,

or gender. These comparisons however reflect that only those variables observed in the data set and cannot be used to indicate whether students choice to not answer the satisfaction question was a reflection of their actual satisfaction. It is not possible to exclude that the possibility using these checks alone, and is addressed further in the next investigation of non-response patterns.

To better the understand this issue, an extra non-response analysis was performed. StudentSurvey.ie uses ten scores based on a cluster of related items from the survey to identify levels of student engagement: Higher-Order Learning (HO), Reflective and Integrative Learning (RI), Quantitative Reasoning (QR), Learning Strategies (LS), Collaborative Learning (CL), Student–Faculty Interaction (SF), Effective Teaching Practices (ET), Quality of Interactions (QI), Supportive Environment (SE) and Learning, Creative and Social Skills (LC). The analytical data for Masters Taught includes 7,188 students of which 5,272 responded to the overall satisfaction question and 1,916 did not.So the students who answered the satisfaction question had on average, 9.79 out of 10 of the engagement indicators completed with 82.7% having completed all 10. In comparison, those who didnt respond to the satisfaction question reported completing an average of only 3.47 indicators. Only 4.5% met all ten indicators; 17.4% met none. The results indicate that there was not a universal response pattern when answering the satisfaction question. Students who completed none of the engagement indicators were likely to have dropped out of the survey before the satisfaction question. The remaining non-respondents are more problematic to the interpret. However A significant number of them completed a large part of the survey, and some respondents did their best to answer the most of the questions to the engagement indicator but did not answer the question on the overall satisfaction. Therefore the lack of response to their question is not likely to be entirely due to the abandonment of the survey and may be due to an intentional refusal to answer that question.

It is a notable that these findings give further support to the absence of a single pattern of behaviour to explain the missing satisfaction responses. They should also be analysed with some care, however, because the position of the overall satisfaction question within the survey is not known. If the question appeared relatively late in the questionnaire, lower completion among students with missing responses could simply reflect the participants leaving the survey before reaching that point, rather than a deliberate decision to skip the satisfaction question. It is not possible to distinguish survey abandonment from the intentional non-response using these data alone. However, when considered alongside the earlier analysis of the observed characteristics of the students with missing responses, the findings suggest that more than one process may have contributed to the missing data.

3.6 Harmonising the 2019 and 2023 Datasets

A series of variables were standardized so that they used consistent names and the category labels across the two survey years (2019 and 2023). Many of the variables studied were measuring the same concepts, but there were still a minor differences in how the variables were named and also how the answers were labelled which made it impossible to combine the variables directly without further preparation. Additionally some categories of the responses

needed to be standardised. The year 2019 study mode category of Part-time/Remote has been recoded to Part-time and added to the year 2023 classification. Similarly, the year 2019 dataset was re-coded with the term International for Non-Irish, to ensure the consistency of terms in the two surveys. The year 2019 data also had an Unknown domicile category, but none were recorded so it was recoded as a missing value. The Age group labels were also harmonised from the 23 years and under to Age 23- and 24 years and over to Age 24+, to match the labelling used in the year 2023 dataset The outcome variable was also being standardised by renaming evalexp (2019) and overall_evaluation (2023) to a shared, common variable name. In the same way, the field of study variable iscedlevel1 has been renamed to isced_level_1 to conform to the naming convention of the year 2023 data set. After these standardisation procedures were being carried out, the two sets of data were merged into a single file with 12,933 student records, and an additional variable was added called "year" to categorise the responses from the year 2019 and year 2023. Following removal of records with a missing value for the outcome variable, the total number of the students used for the modelling was 9457 students, for all further analyses.

For this outcome, the data for the year 2019 was labeled as 'evalexp' and the data for the year 2023 as 'overall_evaluation'. The same applies to the field of study variable with the name iscedlevel1 for the year 2019 and isced_level_1 for the year 2023. Once imported to R there were some changes made to both variables making them common names for analysis as common names within a single dataset.

In the process of preparing the data, two technical problems came through, which needed some modification of the coding strategy. The first is after the MASS package was loaded when doing an ordinal regression analysis as a supplement. Since both MASS and dplyr contain a function named as select(), loading the MASS after dplyr caused its version of the function to mask the dplyr::select() function. This posed the potential for a failure or the unexpected behavior of the select() calls based on the package loading order. To prevent this,so all variable selection operation were written using the fully qualified dplyr::select() syntax, to ensure that a given function is always selected, regardless of the loading order of the packages.Another problem encountered was that while filtering the year 2019 data set, which had been imported from the SPSS file. Variables read into R with read_sav() in the haven package are stored as haven_labelled objects, and thus keep their SPSS labels intact. While these labels are helpful to retain some of the useful metadata, they may result in unusual filtering behaviour when using dplyr::filter()(Wickham, Miller and Smith, 2025). The subset needed was generated by the base R indexing, not filter(), to overcome this problem. The students in the Masters Taught programmes (coded as 5 in PROGTY_RECODED) were selected by means of the square bracket subsetting. This approach yielded the desired subset without compromising for compatibility issues with labelled variables

4 Exploratory Data Analysis

The overall pattern of the satisfaction across the Masters Taught cohort provides important context for the comparison between the Irish and the international students.The data on this level indicated a very little changed between the two survey years 2019 and year 2023. The

overall education experience, proportion of students who reported a positive (81.5%), was similar to the year 2019 (81.2%). This general trend on its own does not imply any of the important differences in the students experiences between the two years. On closer inspection of the overall results, the pattern of results by the domicile is not evident in the headline results. In the year 2019, the Irish students were notably more satisfied than the international students, while in the year 2023 the two rates were similar (Figure 3). The descriptive results presented here do not provide an explanation for the change in the difference between the two groups over time. They do indicate, however that the relationship of the domicile to the overall satisfaction was not consistent in year 2019 and year 2023. The pattern as described here is thus the basis for the more detailed analyses that are performed in the other parts of the study

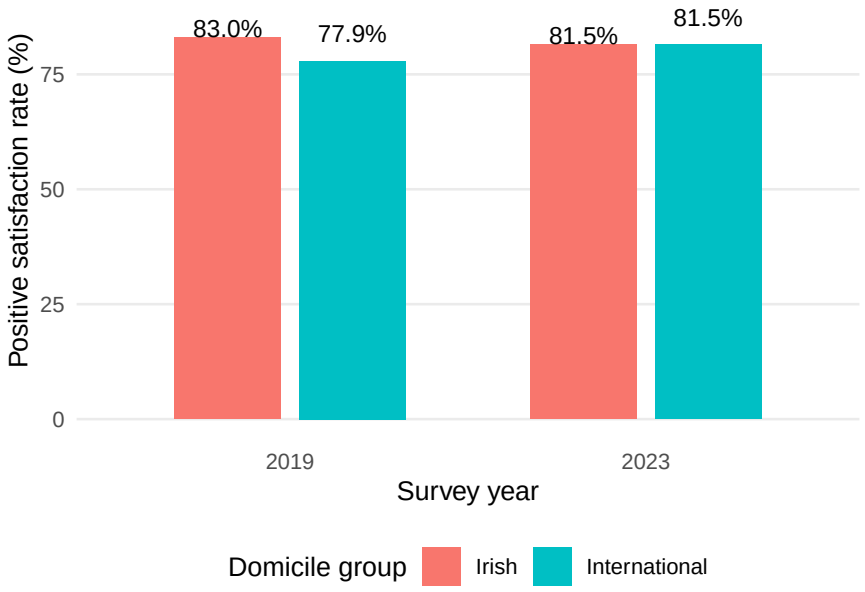


Figure 3: Positive satisfaction rates by domicile group, 2019 and 2023.

The cross tabulation of the age by domicile provides a pattern that generally follows what is expected, but the differences between the groups are relatively small. Figure 4 shows that the international students were slightly more likely to be in the older age group, though the majority of students in both the domicile groups were aged 24 years or older. This indicates that the results found in terms of overall satisfaction between the Irish students and the international students are unlikely to be due to differences in age alone. The association between the age and the domicile runs in the expected direction, but the magnitude is not great enough to treat them as interchangeable. The age variable was thus kept as an independent variable for the analyses that follow

Figure 5 presents the full four-category distribution of satisfaction ratings for each field of study, rather than the positive/negative measure used elsewhere in this chapter. The differences in the number of students choosing "Excellent" are highlighted in this presentation, where they would not be apparent if responses were collapsed into the overall Good/Excellent

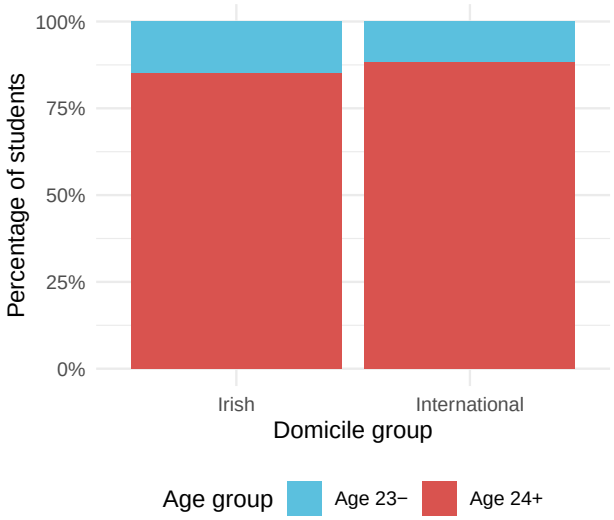


Figure 4: Age profile by domicile group, 2023.

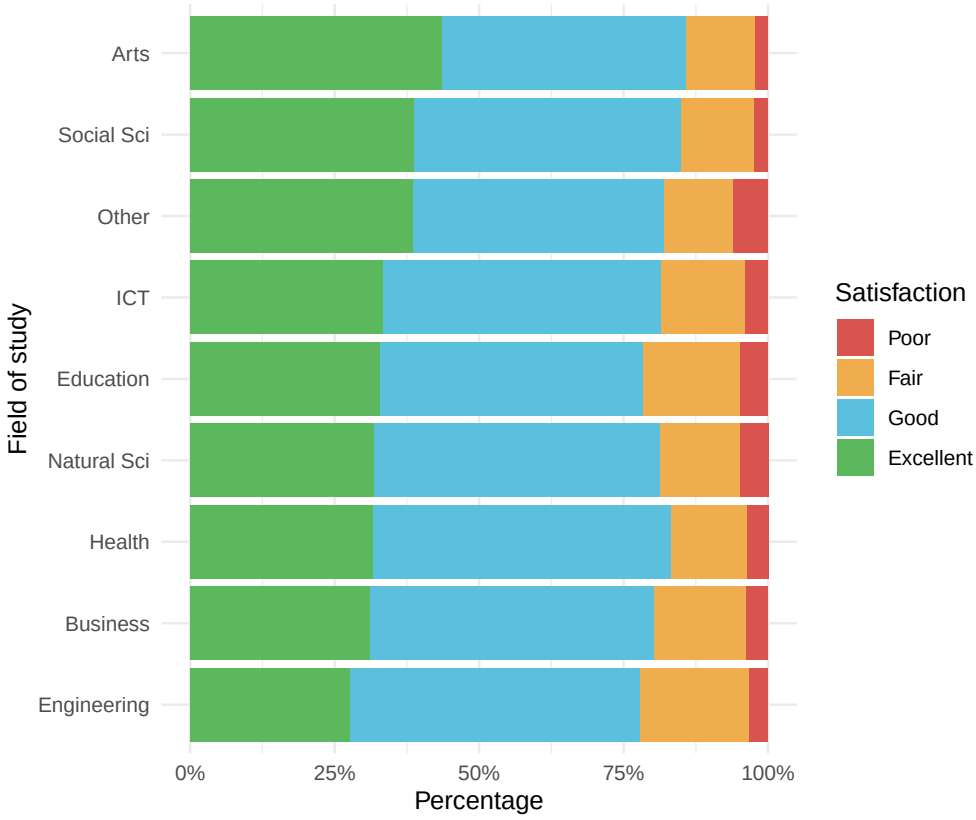


Figure 5: Distribution of satisfaction ratings by detailed field of study, ordered by percentage rating Excellent, 2023.

scale used for the rest of the analyses. As shown in Figure 5, the percentage of the students with an "Excellent" experience differs across disciplines, with Arts having the highest percentage and Engineering the lowest, a difference of 16.0 percentage points between the two. These results are only for the highest satisfaction category and should not be interpreted as the overall positive satisfaction measure, which is based on combining the Good and Excellent responses and is used throughout the rest of this dissertation

It is also noteworthy that the size of the Other category is relatively small. It is not a single field of study, but a combination of Services (n = 76), Generic Programmes (n = 4) and the Agriculture (n = 3). For the Agriculture category, in particular, it was decided not to model it separately because of its small size. If only a few students are represented in a field, the estimates are going to be very volatile and will not be able to be modeled easily with a regression.

This organization was necessary so that there were adequate observations for each category to allow for reliable estimation of the regression model.

When the two variables are analyzed together, there is a pattern in the study mode which is not evident when a each variable is being analyzed individually. Figure 6 shows that the international students reported higher satisfaction overall than the Irish students in the full-time group (n = 2,632 and n = 1,446, respectively). The pattern was the opposite, however in the part-time group, where the Irish students reported higher satisfaction than the international students (n = 1,051 and n = 143, respectively). The difference is not only that the magnitude of the gap varies by the study mode but also that its direction is reversed, such that the group with the highest satisfaction in full-time study reported the lowest satisfaction in part-time study. This reversal by the study mode is a pattern observed in the data rather than evidence of a causal relationship.The Full-time and part-time students may differ in ways that are not captured by this survey, meaning the observed differences in satisfaction cannot be attributed to study mode alone. One additional point should be considered when interpreting the part-time results. With only 143 respondents, the international part-time estimate carries more uncertainty than the estimates for the other three groups.

When the domicile is being analyzed within the large field-of-study groups, this positive relation with positive satisfaction is not uniform across the disciplines. Figure 7 shows that, in the Business group,the Irish students (n = 520) had a slightly higher positive satisfaction rate than the international students (n = 1,040). There was no difference between the Irish (n = 403) and the international students (n = 254) in the Health group. Likewise there was minimal difference in the satisfaction rate of the Rest category, between international students (n = 506) and Irish students (n = 1,095).

A different pattern emerged within the STEM group. International students (n = 975) did better than the Irish students (n = 479), a difference of 6.1 percentage points. This was the greatest difference in the domicile across all four field groupings, compared with the relatively small Irish advantage in Business.

No single pattern by domicile emerges across these four fields. Business and STEM show differences in opposite directions, while Health and the Rest category show little or no difference. The size and direction of the gap appear to vary by field of study rather than by domicile alone.

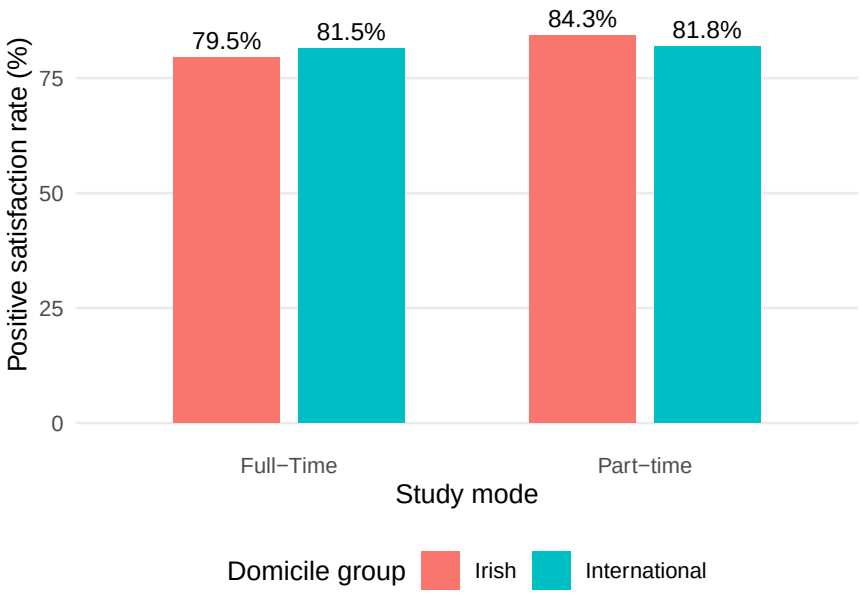


Figure 6: Positive satisfaction rate by domicile, within each study mode, 2023.

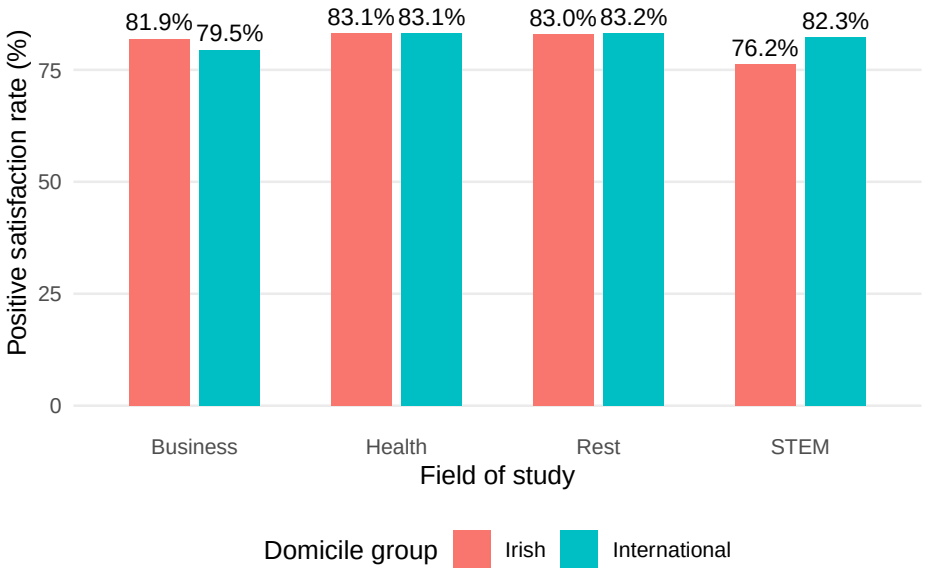


Figure 7: Positive satisfaction rate by domicile, within each field of study, 2023.

The fact that the two domicile groups are quite different when it comes to the living arrangement is evident at just a quick glance at where they reside. The majority of the international students lived in rented homes (38.7%) and Not specified (36.8%), making over three-quarters of the international student body from these two categories. Students from Ireland, on the other hand, distributed fairly evenly among the residence categories. Not specified (27.1%), rented accommodation (25.8%), parental home (23.5%) and own home (18.0%) each contributed a significant proportion. The biggest difference between the two groups was that for those students who resided in their parents home. The difference between the percentages of the international students in this category and the Irish students is 19.7 percentage points (3.8% for the international students, 23.5% for the Irish students). This was also the case for the students who lived in their own house, with only 1.6% of the international students but 18.0% of the Irish students living there, a difference of 16.4 percentage points. Another characteristic of the data that should be noted is the large amount of data in the Not specified category. It represents over one-third of all the international students and just over one-quarter of students in the Ireland, meaning that the residence information is unavailable for a considerable share of the sample.

Figure 8 shows that the international students were more satisfied than the Irish students in the Rented residence category, though the pattern was not consistent across the different accommodation types. Among students who resided with their parents, a similar but smaller gap was found, with the international students reporting higher satisfaction than the Irish students. It was a different story for all the other housing types: Irish students reported higher levels of positive satisfaction than the international students among those living in their own home and in college accommodation.

However these differences must be taken into consideration alongside the underlying sample sizes. The number of Irish own-home students was 461, while the number of international students in that category was 42. The same imbalance existed for students living in college accommodation, with 390 international students and 98 Irish students. The largest gap was in the Other residence category, with international students (n = 146) reporting higher satisfaction than Irish students (n = 36) — the smallest sample size in the table, and therefore the most vulnerable to random variation. The clearest illustration of this is the positive satisfaction rate among Irish students living in college accommodation, based on only 98 students, which may reflect the circumstances of that relatively small subgroup rather than a broader pattern.

Little difference appeared for the students whose home address was categorized as Not specified. We see a broadly similar experiences across this group, with the Irish students (n = 681) and the international students (n = 1,015) reporting comparable rates of positive satisfaction.

Togetherly, these descriptive results show that no single consistent relationship between the residence and the overall satisfaction. Instead, both the direction and the size of the differences vary across accommodation types. This study is exploratory in nature for this reason. The Residence was not included in the regression models presented in the next chapter so the descriptive comparisons provided represent the maximum analysis performed for this variable - without further multivariable modelling to confirm whether the observed differences persist

after the other factors are included.

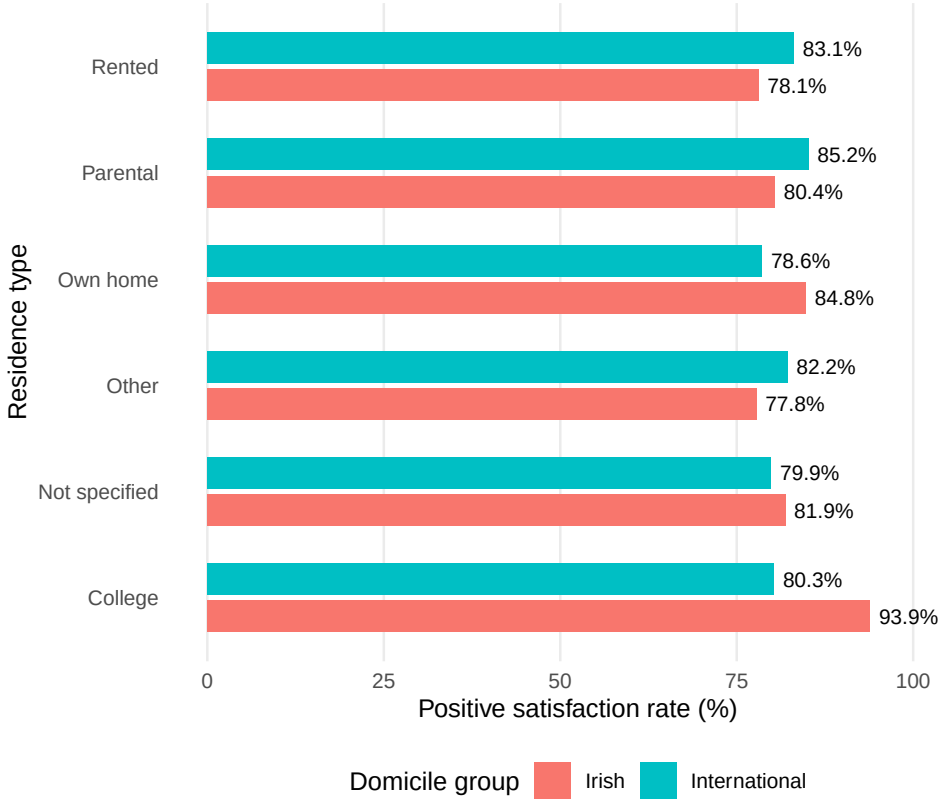


Figure 8: Positive satisfaction rate by domicile, within each residence type, 2023.

5 Analysis and Modelling

5.1 Model Building

Unless otherwise stated the analyses in Sections 5.1 to 5.3 are based on the 2023 dataset only; the combined 2019 and 2023 dataset is introduced separately in Section 5.4.The simplest logistic regression model was used as a starting point in building the models, with the only predictor being the domicile in the overall satisfaction model. This first model had an AIC of 5,055.4 and served as a standard for comparison with all following models. The next step was to test the model to see if demographic data could be added to improve the model. The AGE and GENDER were thus added but this raised the AIC to 5,058.7. The additional complexity of the model did not explain any extra variation in the satisfaction, so demographic variables were not retained in this model.

A more useful improvement when study mode and field of study was introduced into the model. The model fit improved meaningfully with these variables, as it resulted in an AIC

of 5,048.7.These academic characteristics accounted for sufficient explanatory power to guarantee their inclusion in the model that this was the baseline main-effects model used in subsequent steps of the development of the model.

Adding the residence to the main-effects model did not improve model fit. Although the residence was considered a plausible influence on the student satisfaction, its inclusion increased the AIC to 5,056.7. The models showed a poorer fit when the residence was added to the model after the inclusion of the domicile, study mode and the field of study, suggesting that the residence did not have a significant effect after these factors were included in the model. It was decided to drop the variable at this stage, rather than making it the subject of more later models.

To determine if the linkage between the domicile and general satisfaction varied by other student variables, the analysis was conducted again with the addition of the interaction terms. The first interaction tested was domicile × age, and this replaced the standalone main effect of the age in the model. The AIC for this model was 5,050.5, which remained higher than the main-effects baseline (AIC = 5,048.7), and the interaction term was not statistically significant (p = 0.626). Combined with the lack of statistical support, the lack of model fit was not compelling enough to keep this interaction around, and it was dropped from later analyses.

The domicile × study mode interaction resulted in a slight improvement in the model fit, AIC=5,048.3. The improvement over the main-effects model, however, was rather small. This suggested that the association between the domicile and the overall satisfaction was not entirely consistent across different modes of study, making this interaction worth carrying forward into the next stage of the model development

Adding the three-way interaction between the domicile, study mode and field of study resulted in the greatest improvement in the models performance. This gave the lowest AIC, 5044.4, of all the candidate models, showing that the data included patterns that these three characteristics were able to capture that were not found in simpler specifications. Based on this, the three-way interaction model was chosen as the main model for the further analysis.

The three-way model was been again tested with residence as an additional factor as a last check. Once Again, residence did not improve the model fit, as an evidenced by an increase in the AIC to this value, 5,052.3, even after the more complex interaction structure had been introduced. This second non-successful test confirmed the first and contributed to the decision (made in the section of methodology) to not include residence in the final model.

Table 1: AIC values and outcomes for all candidate models fitted on the 2023 dataset.

Model	Predictors	AIC	Outcome
1	Domicile only	5055.4	Baseline
2	Domicile + age + gender	5058.7	Rejected: AIC increased
3	Domicile + age + gender + study mode + field of study	5048.7	Selected: main-effects baseline

Model	Predictors	AIC	Outcome
3b	Model 3 + residence	5056.7	Rejected: AIC increased
4	Domicile x age + gender + study mode + field of study	5050.5	Rejected: AIC increased, p = 0.626
5	Domicile x study mode + age + gender + field of study	5048.3	Retained: small AIC improvement
6	Age + gender + domicile x study mode x field of study (three-way)	5044.4	Selected: PRIMARY MODEL
6b	Model 6 + residence	5052.3	Rejected: AIC increased

5.2 Model 6 Results

Model 6 can be expressed formally as:

$$\text{logit}(\pi) = \beta_0 + \beta_1(\text{Age}) + \sum_{k=1}^{2} \beta_{2,k}(\text{Gender}_k) + \beta_3(\text{Domicile}) + \beta_4(\text{StudyMode}) + \sum_{k=1}^{3} \beta_{5,k}(\text{ISCED}_k)$$

$$+ \beta_6(\text{Domicile} \times \text{StudyMode}) + \sum_{k=1}^{3} \beta_{7,k}(\text{Domicile} \times \text{ISCED}_k) + \sum_{k=1}^{3} \beta_{8,k}(\text{StudyMode} \times \text{ISCED}_k)$$

$$+ \sum_{k=1}^{3} \beta_{9,k}(\text{Domicile} \times \text{StudyMode} \times \text{ISCED}_k)$$

where π is the probability of the reporting a positive satisfaction; β_0 is the intercept; and β_1 through β_9 represent the coefficients (or vectors of coefficients, for categorical predictors with more than two levels) associated with each main effect and interaction term. The Domicile, StudyMode, and Age are binary and it contribute a single coefficient each, with reference categories of the Irish, Full-Time, and Age 23- respectively; Gender contributes two coefficients for its three levels (Female, Male, and Undeclared), with Female as the reference category, and ISCED contributes three coefficients (reference: Business) for its four levels. Interaction terms involving ISCED therefore expand accordingly (e.g., β_7 represents three coefficients for Domicile×ISCED).

A serial analysis of deviance was used to determine the contribution of each term of the final model. It is different from the tests that test each predictor in isolation, since it sequentially adds the predictors to the model in a fixed order, and asks whether each new predictor when added to the model provides a statistically meaningful improvement in the models fit, after the results of the predictors added in previous steps are taken into the account. This means that the importance of any one term will be influenced by not just how much it correlates with the outcome, but it also by the position of the term in the model. Table 2 presents the sequential analysis of deviance for Model 6.

Table 2: Sequential analysis of deviance, Model 6.

Term	Deviance	df	p-value
Age	0.654	1	0.419
Gender	2.068	2	0.356
Domicile	0.005	1	0.945
Study mode	10.095	1	0.001
Field of study (ISCED)	7.924	3	0.048
Domicile x Study mode	2.403	1	0.121
Domicile x ISCED	7.872	3	0.049
Study mode x ISCED	7.014	3	0.071
Domicile x Study mode x ISCED	7.037	3	0.071

The Age and gender were the initial variables that were evaluated the following intercept. None of these variables reached the traditional threshold for the significance ($p < 0.05$). This indicates that at this point in the modelling process, there was little evidence that the age or gender differences affected the students overall satisfaction scores. Nevertheless both the variables were included in the analysis as the standard demographic control variables and were compared later based on these variables without being affected by the significance of these variables. However when analyzed as a primary effect, the domicile was not associated with the overall satisfaction. This was the most obvious evidence that there was no domicile effect on the satisfaction levels. That is, the overall satisfaction was similar across the Irish and the international students before taking into account any interactions with the other variables.This result is obtained from the sequential (Type I) analysis of deviance as seen in Table 2, where each term is evaluated sequentially adding it to the model. The effect of the domicile is assessed prior to the model including the interaction terms since the domicile appearance takes priority over the appearance of study mode, field of study, and interaction terms. The p-value of 0.945 is therefore the p-value due to the independent main effect of domicile only, not affected by the interaction terms added later in the modelling process. It is therefore a direct test to see if there is any association between domicile and overall satisfaction. This discovery was a strong early sign that any differences that might be found were as much a function of other aspects of students experiences as it was a function of domicile alone.

The pattern changed once academic characteristics entered the model. Study mode made a statistically significant contribution, giving the strong evidence that there was a difference in satisfaction between the modes of study. The second variable to reach statistical significance, even though with less evidence, was the field of study, just below of the 5% mark for statistical significance. While these results taken together it indicated that the academic context could better account for the overall satisfaction than the demographic variables used earlier in the modelling process.

The interaction terms produced more variable results. The interaction between the domicile and the field of study was statistically significant, meaning that the relationship between the domicile and the overall satisfaction was non-uniform across the various fields of study. In

other words, there seemed to be some differences between the Irish and the international students, which were different across the different subjects in which they studied. The domicile $\times$ study mode interaction and the study mode $\times$ field of study interaction were not found to be significant when the other terms were added to the model. Finally, the full three-way interaction between the domicile, study mode and field of study was the last term evaluated. Although this did not meet the conventional threshold for statistical significance, it was close enough to suggest that a higher-order interaction might be present. This p-value was not the only reason for choosing to keep the 3-way interaction model. Instead, it represented the entire modeling-building process, in which the three-way specification had the smallest AIC of all the candidate models, and was also favored by the previous likelihood ratio test of the interaction model against the simpler baseline model. Overall these findings were adequate to support the selection of the three-way interaction model as the best model for the subsequent analysis

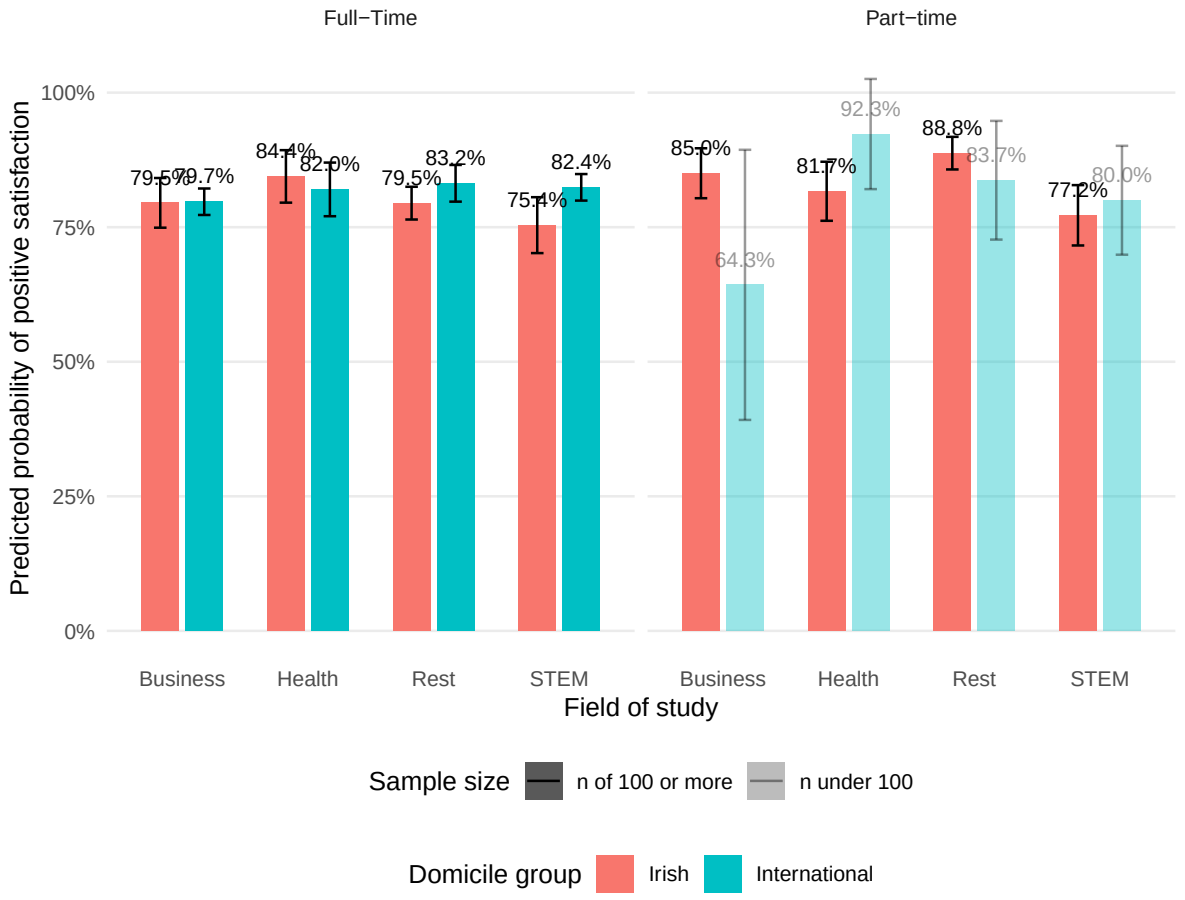


Figure 9: Predicted probability of positive satisfaction by domicile, study mode, and field of study, estimated from the primary three-way interaction model. Bars shown with reduced opacity indicate a cell with fewer than 100 students.

The Final 3-way interaction model was used to estimate the probability of reporting the good overall satisfaction for all possible combinations of the domicile, mode of study and

field of study, as shown in Figure 9. Because the age and the gender were not included in the grouping variables used to generate these predictions, each students predicted probability was calculated using their own actual age and gender, and then averaged within each domicile, study mode, and field of study combination — so the figure reflects the observed mix of ages and genders within each group, rather than a prediction for a single hypothetical student. The figure presents no one consistent distinction between the Irish and the international students but instead shows an interaction between the domicile and satisfaction in different contexts of students academic studies. This visual pattern is similar to the previous modelling results, in which the interaction terms showed that the effect of domestic status on satisfaction was not fixed, but depended on the specific combination of study mode and field of study. The most consistent and evident trend is in the STEM programmes, with a difference of 7.0 percentage points between the international and the Irish students among full-time students. This comparison is based on substantial numbers of students in both groups (International n = 915; Irish n = 264), making it one of the most reliable contrasts observed in the figure. The same direction of difference was also observed in part-time STEM students. The international part-time group was smaller (n = 60), however the trends were comparable across both modes of study with STEM the only subject area where both modes of study predicted higher satisfaction for the international students than for the Irish students.

For many of these combinations, there were no meaningful differences between the domicile groups. In cases such as this, there was little difference in the predicted satisfaction of the full-time Business students, with probabilities of 79.7% for the international students and 79.5% for the Irish students. This near-equivalence bolsters the overall result that the domicile was not linked to satisfaction in a consistent manner throughout the data. Instead, whether a difference emerged was contingent on the specific academic settings in which students learned. The biggest apparent differences are in the part-time Business and part-time Health groups, but both rest on very small international samples of just 14 and 26 students, respectively. In part-time Business, Irish students were predicted to have higher satisfaction than international students (85.0% versus 64.3%). In part-time Health, the pattern was reversed, with international students predicted to have higher satisfaction than Irish students (92.3% versus 81.7%). These differences are apparent in the figure, but with so few international students in either group they are better treated as tentative observations than as evidence of genuine differences between the groups.

In a general, the figure confirms the results of the regression modelling Predicted probabilities vary noticeably across the sixteen combinations of the domicile, study mode and the field of study, indicating that the relationship is context-dependent rather than constant. These academic characteristics need to be viewed in combination to make the relationship apparent, which further prove the three-way interaction model used.

5.3 Model Diagnostics

The Hosmer-Lemeshow goodness of fit test was been used for the evaluating the model calibration (Hosmer, Lemeshow and Sturdivant, 2013). The test compares the probabilities predicted by the model with the outcomes actually observed by the students, with the students

broken up into ten groups according to their predicted probability of reporting a positive satisfaction. The number of students who reported being satisfied is compared to the number of students that would be expected to be satisfied, in each group. The chi-square value obtained from the test was 4.034 with 8 degrees of freedom which resulted in p value of 0.854. The p-value is large, meaning that there is no statistically significant difference between the observed and predicted values across the 10 groups, and therefore good calibration in the model. This indicates that, among groups of students the model assigned similar predicted probabilities to, the actual proportion reporting positive satisfaction closely matched the models expected probabilities. The Area Under the Receiver Operating Characteristic Curve (AUC) was used to evaluate the model's performance in discriminating between students who reported positive satisfaction and those who reported negative satisfaction. This model had an AUC of 0.565. A score of 0.50 means that the discriminatory ability is at chance level, and a score of 1.00 implies perfect discriminatory ability. The value reported here is marginally higher than chance, however indicating that the model does slightly better than random classification, but does not have great ability to distinguish between the individual students who report a positive satisfaction and those who do not

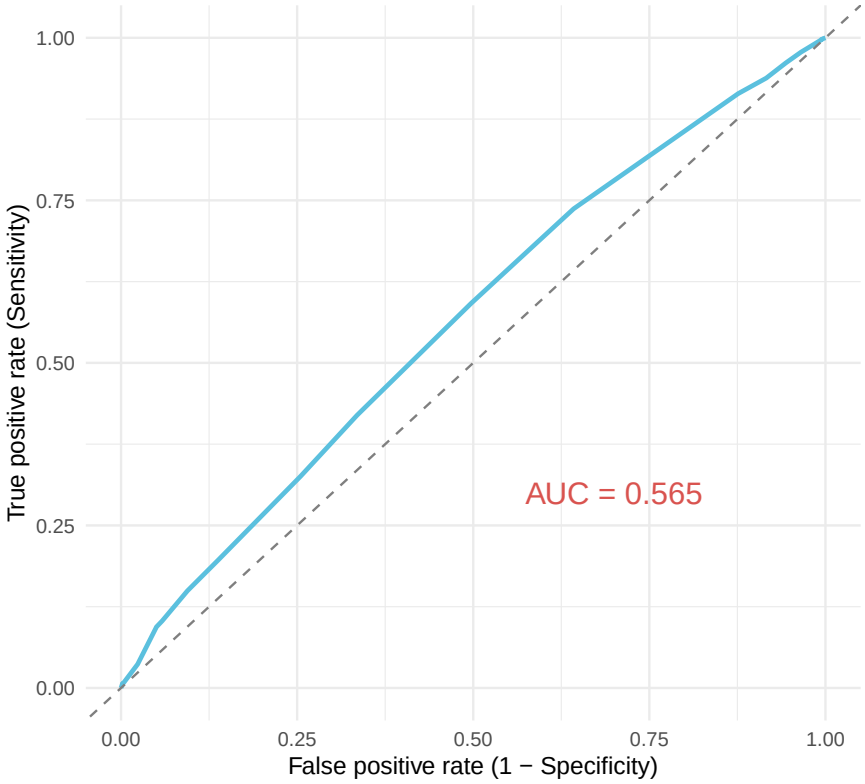


Figure 10: ROC curve for the primary model (Model 6), fitted on the 2023 dataset.

Both are diagnostic measures of the model, and should be thought of as complementary rather than competing. So The Hosmer-Lemeshow test suggests that the predicted probabilities are well calibrated, meaning that the models probability estimates are close to the observed results for groups of the students with similar predicted probability. The AUC, on the other hand, examines the performance of such predictions at the individual level, measuring

the degree to which they separate the students who are satisfied from those who are been dissatisfied. A model can be well calibrated while still having only modest discriminatory ability, and that is the pattern observed here.

However the small AUC cannot be interpreted as a failure of the model, or a poor performance of the model. It was not the goal of this analysis to forecast an individual students positive satisfaction. Rather the model was designed to explore the possibility of differences in the overall satisfaction at the population level, as a function of the interaction between the domicile and mode and field of study. These factors contributed significantly to how satisfied people were, in the earlier likelihood ratio tests and in the sequential analysis of deviance. Many other factors which do not appear in this data set influence an individual students experience, including expectations, relationship with the lecturers, programme-specific experiences and wider personal factors. It is not surprising that there is some individual discrimination in the model, since it only contains a small number of academic and the demographic variables. The results of the diagnostics do not detract from the findings of the study, but they show that the model is calibrated in an appropriate way and that the model fulfills the object of the study, which is to explain, not predict.

5.4 Temporal Comparison

Two likelihood ratio tests were performed to determine whether there was a change in overall satisfaction from the year 2019 to year 2023. Both the tests took into account the survey years, but they were different tests that answered different questions, and thus complementary. Reporting both is important because the first will determine if there was a general change over the two survey years and the second will determine if the change was different for the Irish students and the international students.The models in the Table 3 are a series of nested models that assess changes in the overall student satisfaction from survey year to survey year. It also drew comparisons to the Irish students satisfaction over time and the international students satisfaction over time to determine whether or not there was a difference in trend between the two groups

Table 3: AIC and likelihood ratio test results for the nested models used to test the effect of survey year, fitted on the combined 2019 and 2023 dataset.

Model	Predictors	AIC	Deviance	df	p-value
A	Age + gender + domicile x study mode x ISCED	9062.1	–	–	–
B	A + year (additive)	9063.4	0.706	1	0.401
C	B + domicile x year (interaction)	9054.6	10.788	1	0.001

The pairings were first for the Model 6 specification (Model A in Table 3), with an otherwise identical model that added survey year as an extra main effect. This was to determine whether survey year affected students overall satisfaction similarly, irrespective of their domicile. The

addition of survey year resulted in a small reduction in deviance (0.706 on 1 d.f.), but the effect was not statistically significant (p = 0.401). This suggests that, when these factors of domicile, study mode, field of study and the interaction between them are taken into account, there was no significant extra variation in overall satisfaction between years of the survey. No evidence was found that an overall year effect existed that could be applied to the entire student population.

A domicile × survey year interaction was then added to examine whether changes in the overall satisfaction between the year 2019 and the year 2023 differed for the Irish and the international students.This enabled the overall satisfaction trends over time to differ between the Irish and the international students, instead of assuming that both groups had the same trend over the course of the four years from 2019 to 2023. This interaction greatly enhanced the model, adding 10.788 deviance to the model on 1 degree of freedom (p = 0.001). The model was also favored based on the Akaike Information Criterion (AIC) which went down from 9,063.4 to 9,054.6.These findings indicate that the relationship between the domicile and the overall satisfaction changed between the two survey years, with the Irish and the international students following different trends over the time.

This difference between these two tests can help to understand why the first test was not significant while the second test was. The additive model doesnt allow for there to be a difference between the Irish and the international students in the year effect. When one group doest change over time and the other group increases over time, averaging the two can give an average which looks close to 0. The interaction model explicitly accounts for this difference in each domicile group by allowing for their trend to vary, and thus provides a trend that is not evident in the simpler additive model.The estimated value of the regression coefficients confirms this interpretation. The coefficient for survey year (2023) was -0.117 (p = 0.111) which corresponds to the change in the log-odds of reporting a positive satisfaction for the Irish students who were used as a reference group in this study. This change was not significant and small, indicating there was no significant change in satisfaction among Irish students over the period 2019 to 2023. The interaction term between the international domicile and survey year, however, had a positive and statistically significant value (β = 0.358, p = 0.001). This reflects the extra change reported by the international students compared to that of the Irish students and suggests that there was a statistically significant increase in positive satisfaction reported by the international students over the two service years.

The satisfaction gap between the Irish and the international students narrowed between the year 2019 and the year 2023, and this difference remained statistically significant after controlling for study mode and field of study. The analysis shows that the change over time was specific to the international students and not shared equally between the two domicile groups. The model establishes that this pattern is statistically significant, but it cannot determine why it emerged. Possible explanations are considered in the concluding chapter.

5.5 Supplementary Modelling

In order to validate the findings of the binary logistic model, in which the overall satisfaction response (four categories) remained constant, an ordinal logistic regression model was specified by the function polr() in the MASS package (Venables and Ripley, 2002) with the same variables. In this process two models were been compared. The first one was based only on the domicile, and it came up with an AIC score of 11,841.5. The second added the age, gender, study mode and the field of study reduced the AIC to 11,814.8. This improvement was confirmed by a likelihood ratio test that revealed that the model with the demographic and academic variables had significantly better fit than a model with only the variable of domicile (LR statistic = 40.744, df = 7, $p < 0.001$). This level of improvement was comparable to that in binary logistic regression, indicating that the overall results from both modelling approaches were the same in terms of the role these additional predictors played in the model. The results of the adjusted ordinal logistic regression model are shown as odds ratios and 95% confidence intervals in Table 4. Reference categories are Domicile as International, age as Age 23-, gender as Female, study mode as Full-Time and field of study as Business.

Table 4: Odds ratios and 95% confidence intervals for the adjusted ordinal logistic regression model (Model 2).

Term	OR	95% CI	p-value
Domicile (Irish)	0.937	0.830-1.057	0.291
Age (24+)	0.835	0.715-0.976	0.023
Gender (Male)	1.000	0.901-1.109	0.995
Gender (Undeclared)	0.758	0.373-1.543	0.445
Study mode (Part-time)	1.386	1.204-1.595	<0.001
ISCED (Health)	1.054	0.886-1.254	0.554
ISCED (Rest)	1.300	1.133-1.491	<0.001
ISCED (STEM)	0.996	0.871-1.140	0.959

Based on the adjusted ordinal model, only age, the study mode and one field of study comparison were statistically significant factors in the overall satisfaction. The Students between 24 and older had lower odds of reporting a high level of satisfaction than the students 23 years old and under (OR = 0.835, p = 0.023). Study mode was significantly associated with the satisfaction category, the part-time students, having a more favourable satisfaction category compared to the full-time students (OR = 1.386, $p < 0.001$). A similar pattern was found when students who studied subjects grouped as Rest were compared with those studying Business programmes, with the former being more likely to report higher satisfaction (OR = 1.300, p = 0.0002). With an ordinal logistic regression, positive coefficients indicate higher odds of being in a more satisfied category, while negative coefficients indicate lower odds of being in a more satisfied category. These effects were in the same direction as the descriptive results reported above, with part-time students reporting the higher satisfaction than the full-time students and the students in the Rest category reporting a higher satisfaction than those studying Business

Some predictors did not make a significant contribution after the other predictors had been controlled. The Domicile was not statistically significant (p = 0.291), nor were either of the gender categories. Similarly, the comparisons of Business with Health and STEM were not statistically significant. The threshold values between the four levels of satisfaction (Poor|Fair, Fair|Good and Good|Excellent) were also provided by the model, though the odds ratios for these values are not reported. This is to be expected because the thresholds are not explanatory variables they only represent the approximate limits between the adjacent outcome categories and thus do not have the same meaning as the regression coefficients for the explanatory variables.

Both the ordinal and the binary modelling approaches produced a similar results, strengthening confidence in the consistency of the findings. For both the analyses the same set of explanatory variables were used, except for the definition of the outcome variable. The four satisfaction categories were collapsed to positive and negative in the binary model, but they were presented as the original ordered categories in the ordinal model. Even though this is a distinction, both models came to the same result on the issue of the domicile. Nor was there a main effect of the domicile in the binary logistic regression (p = 0.945), nor in the ordinal logistic regression (p = 0.291). The agreement between these two modelling approaches also gives a confidence that no specific relationship between the students overall satisfaction with their college and the domicile could be detected, only when other academic characteristics were included.

The proportional odds assumption is important for the ordinal logistic regression model. (McCullagh, 1980) This is assuming that the relationship between each predictor and the outcome is essentially the same for all ordered splits of the response categories. In practical terms, the effect of a predictor should be similar whether the model is comparing the students who reported at least Fair rather than Poor, at least Good rather than Fair or below, or Excellent rather than all lower categories . If the coefficient is similar for each of these thresholds, then an overall odds ratio can be used to summarize the effect of a predictor. If the coefficient is significantly different, however, the one estimate does not fully reflect the relationship.The threshold-specific coefficients clearly indicate that this assumption was not completely met in the present analysis. The values of some predictors were maintained roughly constant for all three thresholds. The coefficient for the ISCED Rest category also experienced relatively small changes (0.259, 0.213 and 0.283), and the coefficient for STEM stayed near zero (-0.006, -0.011 and 0.001). The relatively stable values indicate that the proportional odds assumption seems acceptable on these predictors.

The worst violation of the assumption was seen in the study mode. The coefficient for it varied from -0.217 at the first threshold to +0.317 at the second and +0.342 at the third. This is not just a change in size but the direction of the association also turns around. For the first threshold, students who said they were Poor satisfied were less likely to be in the higher threshold category than those who said they were Fair or better satisfied, when the part-time study group was compared. However, in the higher thresholds, part-time students had increased odds for more favourable satisfaction outcomes. These patterns are so different across the thresholds that a single overall coefficient is bound to average these patterns together and is not able to capture either of these patterns completely.

The assumption was not completely justified in the studys main variable, the domicile. For the coefficient, it had changed from -0.192 at the first threshold to -0.193 at the second threshold, and then to 0.010 at the last threshold. While this is not as strong as the variation found with the study mode, it does show that the relationship between domicile and overall satisfaction was not completely consistent throughout all of the ordered response categories. The same was true for age, which showed coefficient values of -0.036, -0.174 and -0.194 for age across the three thresholds, indicating a greater impact of age after the first cumulative split. Taken togetherly, the evidence indicates that the proportional odds assumption was not met. The departures are not evenly distributed across the model that they tend to occur in specific predictors, and the example of study mode is quite clear, and the domicile and the age also vary significantly across thresholds. The results corroborate the modelling approach taken in the previous part of the dissertation in which the use of binary logistic regression as the main method of analysis was chosen. The binary logistic regression approach does not rely on the proportional odds assumption and therefore provides a more suitable framework for investigating the relationship between domicile, study mode, field of study and overall student satisfaction. while the ordinal model serves as a supplementary analysis to assess the robustness of the main findings

6 Conclusion

6.1 Summary of Findings

The analysis revealed a different relationship between the domicile and the overall satisfaction in the two periods of the survey.The Irish students reported an overall greater level of positive satisfaction in 2019 compared to the international students. The gap between these two groups was, however it had closed by 2023 when the satisfaction was similar. This convergence is also not just a result of changes in the mix of students in the cohort, as shown in the additional modelling in Table 3. The relationship between the domicile and the overall satisfaction was still statistically significant after the study mode and field of study were accounted for, confirming that the change was genuine rather than an artefact of these other factors.The modelling enables understanding of where this change occurred. Satisfaction among Irish students remained largely consistent between 2019 and 2023, while international students showed a genuine improvement in satisfaction over the same period. The result was a closing of the satisfaction gap, largely due to this increase in satisfaction among international students, and not to any real change among the Irish students

This pattern is only revealed when the interaction between the domicile and survey year is added to the model. A more simple model, which estimated a single year change for all students, would not capture this change as it would assume that both the Irish and the international students followed the same trend over time. In reality, the comparatively stable satisfaction rates of Irish students balanced out the rise in satisfaction for the international students, resulting in few indications of overall satisfaction change when combined.

The results need to be interpreted in the light of the significant changes in the composition of the student population between the two survey years described in the Introduction. The

results of the analysis, however, suggest that these compositional shifts are not the only factors which led to the converging satisfaction gap. Even after adjusting for study mode and field of study, the satisfaction scores of the international students still show a genuine improvement between 2019 and 2023. The models offer strong evidence for this improvement, but they have no explanation as to why it took place. This may be due to a shift in the profile of the students who opt to enroll on Masters Taught programmes between 2019 and 2023. However some factors in the wider economic landscape can play a role in decisions to pursue further study, and this can differ significantly over time - such as graduate employment opportunities. Where the profile of the students entering these programmes has changed between the two survey years, some of the improvement seen among international students may be due to a change in who chose to enter the programmes rather than to changes in the experience of similar students. Other factors, including changes in institutional support, changes in student expectations, recovery from the pandemic or broader social and economic shifts, may have played a role. But the available data do not permit a testing of these explanations or a separation of the explanations themselves.

Study mode and field of study as determined by this research showed the highest structural factors among the students satisfaction. Whereas at the descriptive level, it was found that part-time students were more satisfied in terms of positive aspects than full-time students, the former accounted for a smaller proportion of students. This trend was confirmed by the regression analysis, and both study mode and field of study statistically significantly contributed to the variance in satisfaction. The overall results indicate that the effect on satisfaction is stronger for the manner of study and the subject of study than for domicile alone. This trend occurred in the descriptive analyses reported in Chapter 4 and was found in the regression models in Chapter 5.

Secondly, found that the domicile alone was not statistically significant in explaining global satisfaction, and thus directly relevant to the studys research question. This indicates that difference in satisfaction cannot be attributed to differences in Irish and international students alone. Rather, the connection between the domicile and satisfaction is linked to the academic setting where students are learning. When study mode and field of study were then taken into account together with domicile a more nuanced picture emerged. In some combinations of study mode and discipline, it was predicted that international students would be more likely than Irish students to report high satisfaction levels. This trend was most pronounced in STEM, while only Business was the only field where the trend was in the other direction. The differences between the two domicile groups for the rest of the combinations were relatively small. These findings indicate that the effect of domicile is not consistent, but rather depends on the study mode and field of study, and so address the research question set in the Introduction to the study, namely, do international students have different levels of satisfaction than the Irish students, and what are the reasons for any differences that may be reported?

6.2 Limitations

This study has some cautions to be noted. Firstly, the overall response rate to the survey was 25.7%, which is an indication that approximately three-quarters of the eligible student population did not participate in the survey. Additionally, in both survey years, about one quarter of the respondents did not report on the question of satisfaction with the global outcome, further reducing the number of respondents available for analysis. Observed data was analysed to provide information on whether the missing values were Missing Completely at Random (MCAR) with respect to the measured characteristics: domicile, type of study, age and gender, and there was little evidence found to suggest otherwise, supporting complete case analysis. However, the special non-response analysis indicated that the non-response was not consistent. Larger portions of the survey were also not completed by some students who did not respond to the satisfaction question, a common feature of surveys. Some, however, provided responses to almost all of the questions but did not complete the overall satisfaction question. This pattern cannot be put down to nothing more than disengagement, since the unobserved satisfaction response cannot be excluded. The evidence that is available, therefore, suggests that it is possible that the missing data resulted from a variety of mechanisms rather than one clearly defined mechanism, and there is uncertainty involved throughout.

A second constraint is that the model is unable to separate those students who reported positive satisfaction from those who did not. The last binary logistic regression model had modest discriminatory ability with an AUC of 0.565. This is not considered to be a limitation because it is explained, as previously mentioned in Section 5.3, rather than predicted. Given the limited number of variables in the survey, a higher degree of prediction of individual responses was not expected.

Another limitation of the study is that a cross sectional design was used. Data was analysed for both the year 2019 and year 2023, but both samples were different groups of students. This implies that there are variations between the two. Any differences between the two years of the survey are not necessarily differences in the experience of the individual student; instead the focus is on population level differences.

Another restriction of the present study is that it is not possible to specify the higher education institution that each student has attended. A students satisfaction can differ from one institution to another due to variations in resources, teaching methods, campus facilities, campus support services and other institutional factors, regardless of the students domicile, mode of study or area of study. If an institutional identifier is not included, these differences cant be explored and explained. Therefore, some of the differences in the models may be due to institutional-level differences and not the characteristics studied in this study.

Lastly, the data collected from StudentSurvey.ie centres on students experience at a specific time within an academic year. Student satisfaction is likely to change during the course of a programme as students are exposed to various modules in the programme, various assessment periods, various support services, and various experiences in their lives. The survey is a 'snapshot' of students' experience, and does not provide a comprehensive view of the evolution of satisfaction throughout the course of their education.

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